

1. PROBLEM DEFINITION & SCENARIO

The traditional manual library management system faces significant operational challenges that impede efficiency and accuracy. Libraries typically manage thousands of books, multiple copies of the same title, diverse member registrations, and complex loan-return cycles. Manual tracking of these operations leads to:

- **Inefficient Book Tracking:** Difficulty in monitoring the real-time availability status of individual book copies
- **Error-Prone Loan Management:** Manual calculation of due dates and late fees prone to human error
- **Limited Reservation System:** No systematic way for members to reserve currently loaned books
- **Inadequate Reporting:** Lack of real-time statistical data on library operations and member activities
- **Security Concerns:** No standardized access control for library staff with different authorization levels

Project Objective

The Advanced Library Management System (ALMS) addresses these challenges through a comprehensive relational database management system (RDBMS) that automates and streamlines library operations. The system provides:

- 1. Automated Book Catalog Management:** Track books by ISBN, titles, authors, publishers, categories, and individual physical copies
- 2. Copy-Level Tracking:** Monitor each physical book copy independently with status indicators (Available, On Loan, Maintenance)
- 3. Intelligent Loan Management:** Automate due date calculation, return processing, and penalty assessment
- 4. Member Penalty System:** Dynamic penalty score tracking with configurable fine rates
- 5. Reservation Queue:** Allow members to reserve books currently on loan
- 6. Role-Based Access Control:** Secure employee authentication with Admin and Staff privilege levels
- 7. Real-Time Reporting:** Dashboard with statistics on active loans, overdue returns, and member activities
- 8. Audit Trail:** Process log for tracking all system operations and administrative actions

2. DATA MODELING USING ENTITY-RELATIONSHIP (ER) MODEL

The Entity-Relationship model forms the foundation of our database design. We developed both a conceptual ER diagram and a detailed relational schema to ensure data integrity and eliminate redundancy through normalization to Third Normal Form (3NF).

2.1 Conceptual ER Diagram

The conceptual model represents the high-level relationships between entities in the library system. Key relationships include:

- **WRITES (1:N)**: One author can write multiple books; multiple books can have one author
- **PUBLISHES (1:N)**: One publisher publishes multiple books
- **CATEGORIZES (1:N)**: Books are classified into categories
- **HAS COPY (1:N)**: One book title can have multiple physical copies
- **INVOLVES (N:M)**: Loans connect members to book copies through a relationship entity
- **INCURS (1:N)**: Fines are associated with specific loans
- **BORROWS (1:N)**: Members create multiple loan records
- **REQUESTS (1:N)**: Members can reserve multiple books

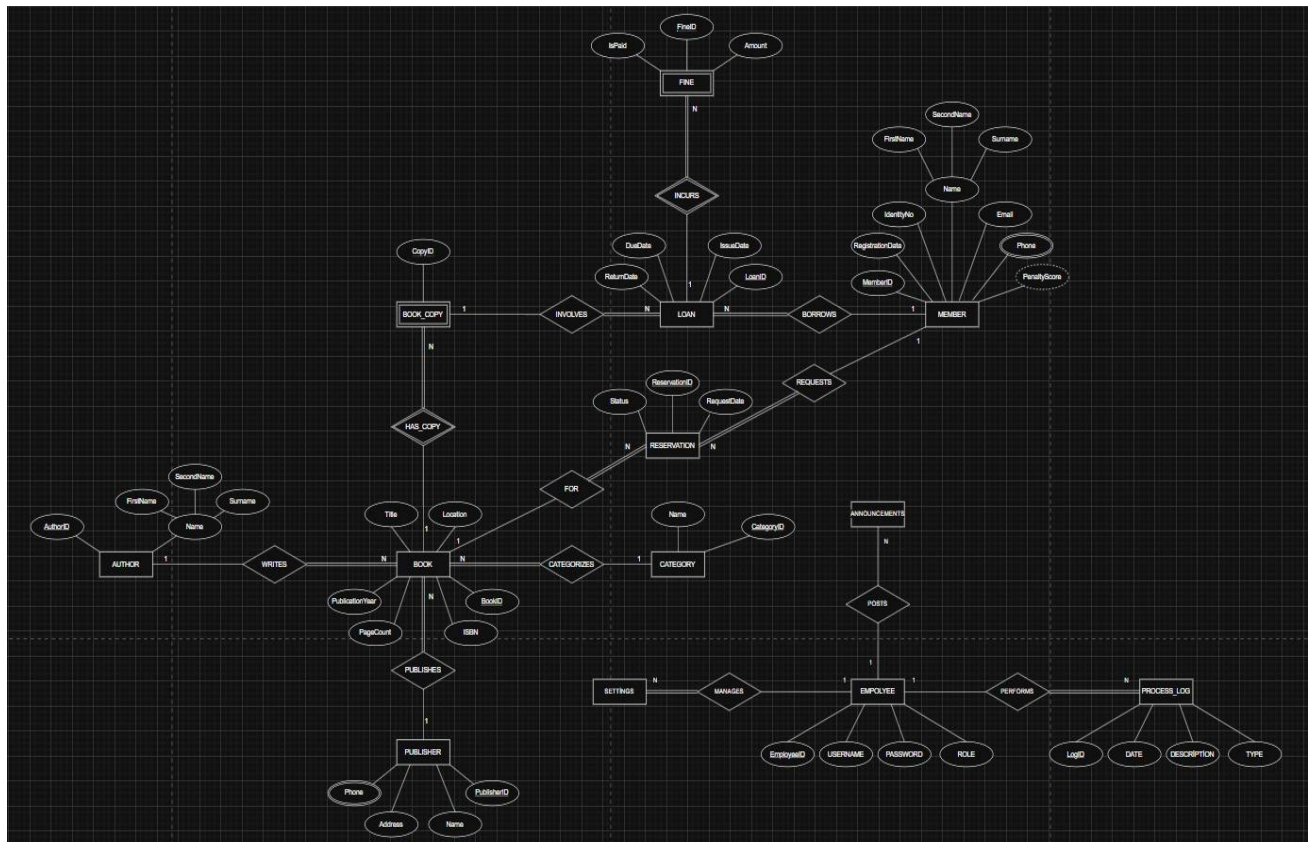


Figure 1: Conceptual Entity-Relationship Diagram

2.2 Relational Schema (Physical ER Diagram)

The relational schema shows the complete database structure with all tables, attributes, primary keys (PK), and foreign keys (FK). This design ensures referential integrity through proper constraint definitions.

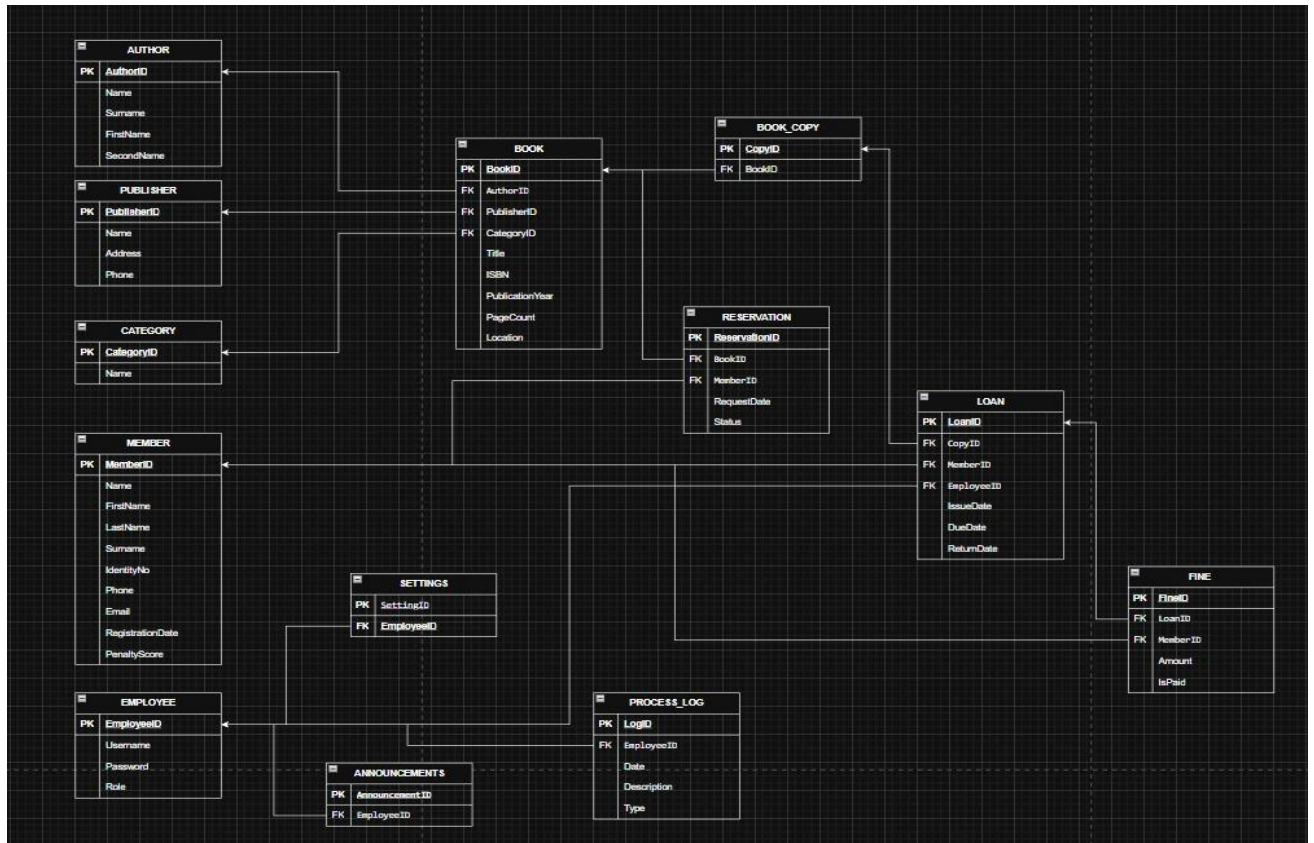


Figure 2: Detailed Relational Schema with Primary and Foreign Keys

3. MAPPING OF RELATIONAL SCHEMA

The transformation from the conceptual ER model to the relational schema follows standard mapping rules. Each entity becomes a table, and relationships are implemented through foreign keys.

3.1 Entity Mapping

Strong Entities:

- AUTHOR → TBL_AUTHORS (AuthorID as PK)
- PUBLISHER → TBL_PUBLISHERS (PublisherID as PK)
- CATEGORY → TBL_CATEGORIES (CategoryID as PK)
- BOOK → TBL_BOOKS (BookID as PK, with FKs to Author, Publisher, Category)
- BOOK_COPY → TBL_BOOK_COPIES (CopyID as PK, BookID as FK)
- MEMBER → TBL_MEMBERS (MemberID as PK)
- EMPLOYEE → TBL_EMPLOYEES (EmployeeID as PK)
- SETTINGS → TBL_SETTINGS (SettingID as PK, EmployeeID as FK)

Weak/Dependent Entities:

- LOAN → TBL_LOANS (LoanID as PK, CopyID and MemberID as FKs)
- FINE → TBL_FINES (FineID as PK, LoanID as FK)
- RESERVATION → TBL_RESERVATIONS (ReservationID as PK, BookID and MemberID as FKs)
- ANNOUNCEMENT → TBL_ANNOUNCEMENTS (AnnouncementID as PK, EmployeeID as FK)
- PROCESS_LOG → TBL_PROCESS_LOGS (LogID as PK, EmployeeID as FK)

3.2 Relationship Mapping

One-to-Many (1:N) Relationships:

- **WRITES:** Implemented by AuthorID foreign key in TBL_BOOKS
- **PUBLISHES:** Implemented by PublisherID foreign key in TBL_BOOKS
- **CATEGORIZES:** Implemented by CategoryID foreign key in TBL_BOOKS
- **HAS COPY:** Implemented by BookID foreign key in TBL_BOOK_COPIES
- **BORROWS:** Implemented by MemberID foreign key in TBL_LOANS
- **INCURS:** Implemented by LoanID foreign key in TBL_FINES

Many-to-Many (N:M) Relationship:

- **INVOLVES:** The relationship between MEMBER and BOOK_COPY through LOAN is resolved using TBL_LOANS as a junction table containing both MemberID and CopyID as foreign keys

4. TABLE DESCRIPTIONS (DATABASE SCHEMA)

The database consists of 13 interconnected tables designed to Third Normal Form (3NF) to ensure data integrity, eliminate redundancy, and optimize query performance.

4.1 TBL_AUTHORS

Purpose: Stores unique author information to prevent redundant text entries in the book catalog.

Column Name	Data Type	Constraints	Description
AuthorID	INT	PRIMARY KEY, <u>IDENTITY</u> (1,1)	Unique identifier for each author
Name	<u>NVARCHAR</u> (100)	NOT NULL	Author's first name
Surname	<u>NVARCHAR</u> (100)	NOT NULL	Author's last name
FirstName	<u>NVARCHAR</u> (50)		Additional first name field
SecondName	<u>NVARCHAR</u> (50)		Middle name or additional name

4.2 TBL_BOOKS

Purpose: Core catalog table storing book titles and metadata with references to authors, publishers, and categories.

Column Name	Data Type	Constraints	Description
PublisherID	INT	PRIMARY KEY, <u>IDENTITY</u> (1,1)	Unique identifier for each publisher
Name	<u>NVARCHAR</u> (200)	NOT NULL	Publisher's official name
Address	<u>NVARCHAR</u> (500)		Publisher's physical address
Phone	<u>NVARCHAR</u> (20)		Contact phone number

4.3 TBL_LOANS

Purpose: Transaction hub recording issue dates, due dates, and return dates. Implements the core borrowing business logic.

Column Name	Data Type	Constraints	Description
CategoryID	INT	PRIMARY KEY, <u>IDENTITY</u> (1,1)	Unique identifier for each category
Name	<u>NVARCHAR</u> (100)	NOT NULL, UNIQUE	Category name (e.g., Sci-Fi, Classic, Horror)

4.4 TBL_BOOKS

Purpose: Core catalog table storing book titles and metadata.

Column Name	Data Type	Constraints	Description
BookID	INT	PRIMARY KEY, <u>IDENTITY(1,1)</u>	Unique identifier for each book title
ISBN	<u>NVARCHAR(20)</u>	UNIQUE	International Standard Book Number
Title	<u>NVARCHAR(300)</u>	NOT NULL	Book title
AuthorID	INT	FOREIGN KEY → TBL_AUTHORS	Reference to author
PublisherID	INT	FOREIGN KEY → TBL_PUBLISHERS	Reference to publisher
CategoryID	INT	FOREIGN KEY → TBL_CATEGORIES	Reference to category
PublicationYear	INT		Year of publication
PageCount	INT		Number of pages
Location	<u>NVARCHAR(50)</u>		Physical shelf location

4.5 TBL_BOOK_COPIES

Purpose: Tracks real-time status and physical location of every individual book unit. This is critical for managing multiple copies of the same title independently.

Column Name	Data Type	Constraints	Description
CopyID	INT	PRIMARY KEY, <u>IDENTITY(1,1)</u>	Unique identifier for each physical copy
BookID	INT	FOREIGN KEY → TBL_BOOKS	Reference to book title
Status	<u>NVARCHAR(20)</u>	NOT NULL, CHECK IN ('Available', 'On Loan', 'Maintenance')	Current availability status

4.6 TBL_MEMBERS

Purpose: Contains member identity records and their dynamic penalty scores.

Column Name	Data Type	Constraints	Description
MemberID	INT	PRIMARY KEY, IDENTITY(1,1)	Unique member identifier
Name	NVARCHAR(100)	NOT NULL	Member's first name
FirstName	NVARCHAR(50)		First name
LastName	NVARCHAR(50)		Last name
Surname	NVARCHAR(100)		Surname
IdentityNo	NVARCHAR(20)	UNIQUE	Government-issued ID number
Phone	NVARCHAR(20)		Contact phone
Email	NVARCHAR(100)		Email address
RegistrationDate	DATETIME	DEFAULT GETDATE()	Registration timestamp
PenaltyScore	INT	DEFAULT 0	Accumulated penalty points

4.7 TBL_LOANS

Purpose: Transaction hub recording issue dates, due dates, and return dates. Implements the core borrowing business logic.

Column Name	Data Type	Constraints	Description
LoanID	INT	PRIMARY KEY, IDENTITY(1,1)	Unique loan transaction ID
CopyID	INT	FOREIGN KEY → TBL_BOOK_COPIES	Reference to borrowed copy
MemberID	INT	FOREIGN KEY → TBL_MEMBERS	Reference to borrowing member
EmployeeID	INT	FOREIGN KEY → TBL_EMPLOYEES	Staff who processed loan
IssueDate	DATETIME	NOT NULL, DEFAULT GETDATE()	Date book was loaned
DueDate	DATETIME	NOT NULL	Expected return date
ReturnDate	DATETIME	NULL	Actual return date (NULL if not returned)

4.8 TBL_FINES

Purpose: Automatically links overdue returns to financial penalty records.

Column Name	Data Type	Constraints	Description
<u>FineID</u>	INT	PRIMARY KEY, <u>IDENTITY</u> (1,1)	Unique fine identifier
<u>LoanID</u>	INT	FOREIGN KEY → TBL_LOANS	Reference to loan that incurred fine
<u>MemberID</u>	INT	FOREIGN KEY → TBL_MEMBERS	Reference to member
<u>Amount</u>	<u>DECIMAL</u> (10,2)	NOT NULL	Fine amount in currency
<u>IsPaid</u>	BIT	DEFAULT 0	Payment status flag

4.9 TBL_RESERVATIONS

Purpose: Manages requests for books currently lent out to other members.

Column Name	Data Type	Constraints	Description
<u>ReservationID</u>	INT	PRIMARY KEY, <u>IDENTITY</u> (1,1)	Unique reservation ID
<u>BookID</u>	INT	FOREIGN KEY → TBL_BOOKS	Reference to book title
<u>MemberID</u>	INT	FOREIGN KEY → TBL_MEMBERS	Reference to reserving member
<u>RequestDate</u>	DATETIME	DEFAULT <u>GETDATE</u> ()	Date reservation was made
<u>Status</u>	<u>NVARCHAR</u> (20)	CHECK IN ('Pending', 'Fulfilled', 'Cancelled')	Reservation status

4.10 TBL_EMPLOYEES

Purpose: Stores staff credentials and system access roles.

Column Name	Data Type	Constraints	Description
<u>EmployeeID</u>	INT	PRIMARY KEY, <u>IDENTITY</u> (1,1)	Unique employee identifier
<u>Username</u>	<u>NVARCHAR</u> (50)	NOT NULL, UNIQUE	Login username
<u>Password</u>	<u>NVARCHAR</u> (256)	NOT NULL	Hashed password
<u>Role</u>	<u>NVARCHAR</u> (20)	CHECK IN ('Admin', 'Staff')	Authorization level

4.11 TBL_ANNOUNCEMENTS

Purpose: Communication tool for staff to notify members of library news and updates.

Column Name	Data Type	Constraints	Description
AnnouncementID	INT	PRIMARY KEY, IDENTITY(1,1)	Unique announcement ID
EmployeeID	INT	FOREIGN KEY → TBL_EMPLOYEES	Staff who posted announcement
Title	NVARCHAR(200)	NOT NULL	Announcement headline
Content	NVARCHAR(MAX)		Full announcement text
Date	DATETIME	DEFAULT GETDATE()	Publication date

4.12 TBL_SETTINGS

Purpose: Dynamic configuration table for system-wide variables like daily fine rates.

Column Name	Data Type	Constraints	Description
SettingID	INT	PRIMARY KEY, IDENTITY(1,1)	Unique setting identifier
EmployeeID	INT	FOREIGN KEY → TBL_EMPLOYEES	Staff who modified setting

Column Name	Data Type	Constraints	Description
DailyFineAmount	DECIMAL(10,2)		Fine amount per day overdue
MaxLoanDuration	INT		Maximum loan period in days
LastModified	DATETIME	DEFAULT GETDATE()	Last update timestamp

4.13 TBL_PROCESS_LOGS

Purpose: Auditing table that logs every administrative action for security and accountability.

Column Name	Data Type	Constraints	Description
LogID	INT	PRIMARY KEY, <u>IDENTITY(1,1)</u>	Unique log entry ID
EmployeeID	INT	FOREIGN KEY → TBL_EMPLOYEES	Staff who performed action
Date	DATETIME	DEFAULT <u>GETDATE()</u>	Action timestamp
Description	<u>NVARCHAR(500)</u>		Description of action
Type	<u>NVARCHAR(50)</u>		Action type (INSERT, UPDATE, DELETE)

5. USER INTERFACE DESIGN & FUNCTIONALITY

The system features a Windows Forms-based C# application providing an intuitive, professional interface for library staff. The UI is organized into functional modules accessible through a sidebar navigation menu.

5.1 Login Interface

Secure authentication system for library staff with role-based access control.

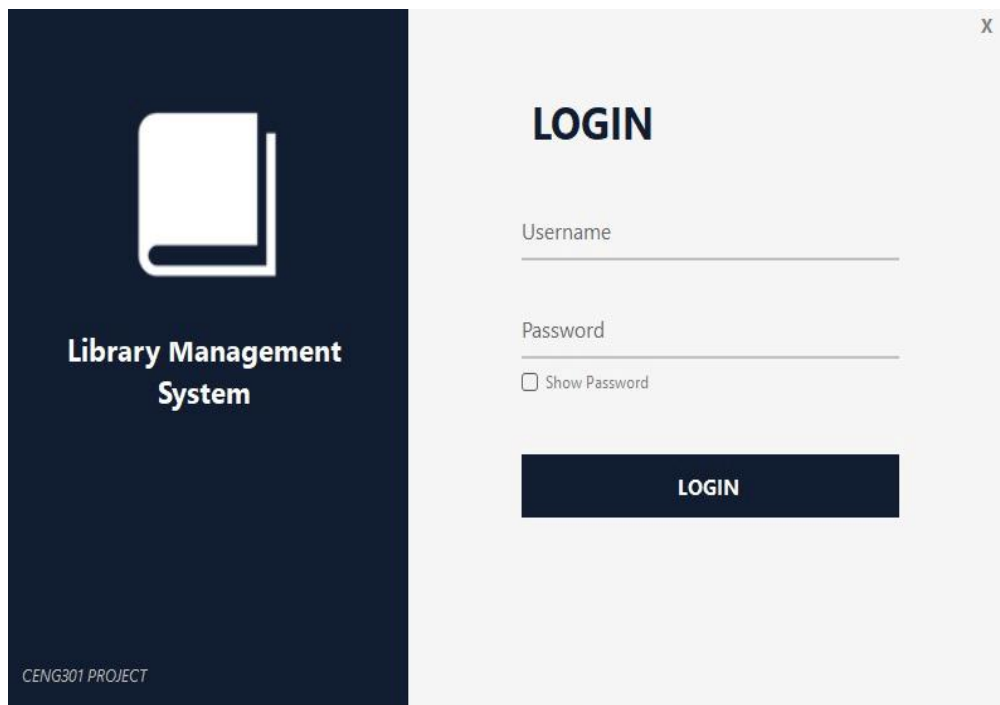


Figure 3: Login Interface

Key Features:

- Username and password authentication with secure credential validation against TBL_EMPLOYEES
- Password visibility toggle for user convenience
- Role-based access control (Admin/Staff roles) determining available features
- Session management with employee ID tracking for audit logs

5.2 Dashboard (Home Screen)

The dashboard provides at-a-glance statistics and real-time library activity monitoring.

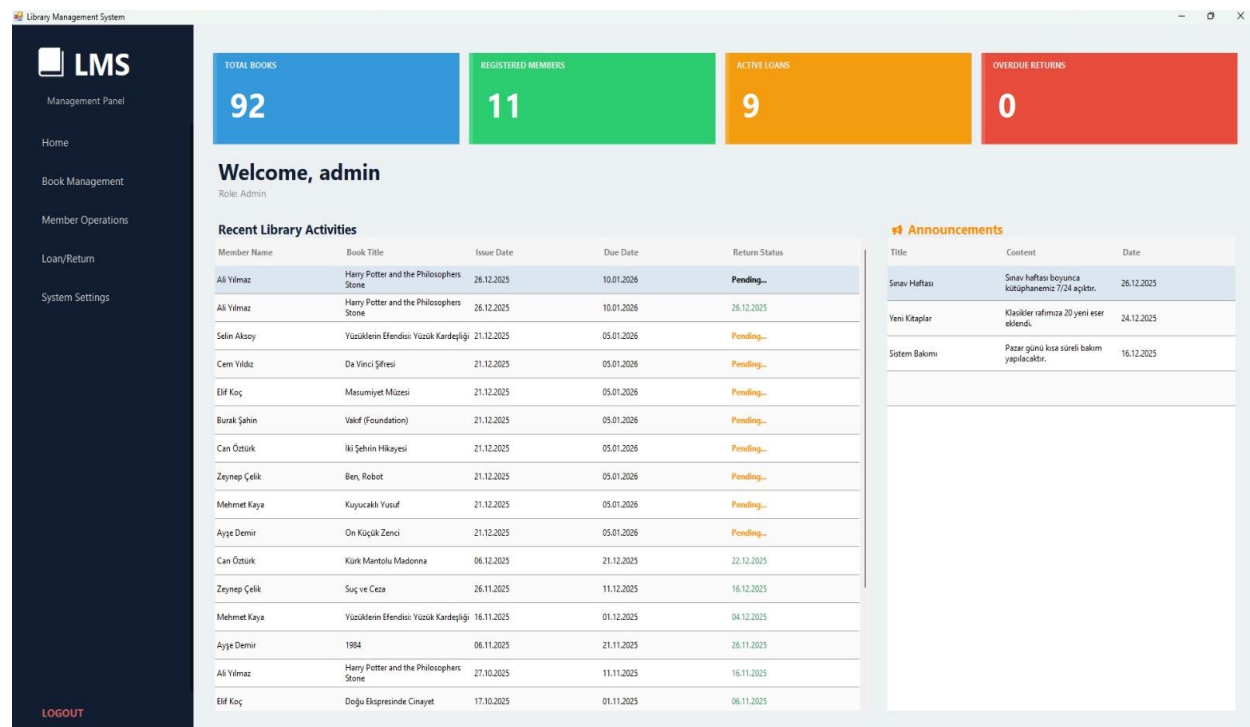


Figure 4: Dashboard showing library statistics and recent activities

Key Features:

- Statistical Cards:** Color-coded metrics showing Total Books (92), Registered Members (11), Active Loans (9), and Overdue Returns (0)
- Recent Library Activities:** Real-time table displaying member names, book titles, issue dates, due dates, and return status with color indicators (Pending in orange, Returned in green)
- Announcements Panel:** Important library notices including 'Sınav Haftası' reminders and system maintenance updates
- Welcome message with admin name and role display

5.3 Book Operations Interface

Comprehensive book catalog management with inventory tracking and copy management.

The interface is divided into three main sections:

- Left Sidebar:** Contains navigation links: Management Panel, Home, Book Management, Member Operations, Loan/Return, and System Settings. A LOGOUT button is at the bottom.
- Top Section: BOOK OPERATIONS**
 - ISBN (Barcode):
 - Book Title:
 - Category: Science Fiction (dropdown with a plus icon)
 - Author: J.K. Rowling (searchable dropdown with a plus icon)
 - Publisher: Penguin Books (dropdown with a plus icon)
 - Pages: Year: Shelf:
 - Buttons: SAVE (green), UPDATE (orange), DELETE (red)
- Right Section: Book Catalog Table**
 - Search bar: + ADD STOCK Selected Book Stock: -
 - Table with columns: ISBN, Book Title, Category, Author, Publisher, Pages, Year, Location, Stock.
 - Buttons: DELETE Selected Copy, Available, Maintenance, Lost

ISBN	Book Title	Category	Author	Publisher	Pages	Year	Location	Stock
9789750030	Dava	Polisiye	Franz Kafka	Oxford University ...	230	1925	Raf H-1	5
9789750029	Anna Karenina	Novel	Lev Tolstoy	Pegasus Yayınları	864	1877	Raf C-8	3
9789750028	Benim Adım Kırmızı	Korku	Orhan Pamuk	İthaki Yayınları	450	1998	Raf F-6	3
9789750027	On Küçük Zencî	Personal Develop...	Agatha Christie	Metis Yayınları	224	1939	Raf D-4	3
9789750026	Medyum	Fantastik	Stephen King	Can Yayınları	500	1977	Raf E-1	3
9789750025	Melekler ve Şeytan	Personal Develop...	Dan Brown	Can Yayınları	520	2000	Raf D-3	3
9789750024	Kuyucaklı Yusuf	Korku	Sabahattin Ali	Penguin Books	220	1937	Raf G-3	3
9789750023	Yaratıldın Nottlar	Klasikler	Fyodor Dostoyevski	Pegasus Yayınları	140	1864	Raf C-7	3
9789750022	Hayvan Çiftliği	Korku	George Orwell	Oxford University ...	112	1945	Raf B-4	3
9789750021	Ben, Robot	Science Fiction	Isaac Asimov	Yapı Kredi Yayınları	253	1950	Raf B-2	3
9789750020	Tutunamayanlar	Korku	Oğuz Atay	İthaki Yayınları	724	1972	Raf G-2	3
9789750019	İmkansızın Şarkısı	Korku	Haruki Murakami	Pegasus Yayınları	350	1987	Raf F-5	3
9789750018	İki Şehrin Hikayesi	Novel	Charles Dickens	Oxford University ...	448	1859	Raf C-6	3
9789750017	Çanlar Kimin İçin ...	Korku	Ernest Hemingway	Pegasus Yayınları	471	1940	Raf F-4	3
9789750016	Gurur ve Önyargı	Korku	Jane Austen	Pegasus Yayınları	432	1813	Raf F-3	3
9789750015	Donüşüm	Novel	Franz Kafka	Oxford University ...	80	1915	Raf C-5	3
9789750014	Savaş ve Barış	Novel	Lev Tolstoy	Pegasus Yayınları	1225	1869	Raf C-4	3
9789750013	Sefiller	Novel	Victor Hugo	Pegasus Yayınları	1463	1862	Raf C-3	3
9789750012	Vakıf (Foundation)	Science Fiction	Isaac Asimov	Yapı Kredi Yayınları	255	1951	Raf B-1	3
9789750011	İnce Memed 1	Korku	Yagar Kemal	Penguin Books	436	1955	Raf G-1	3
9789750010	Aşk	Korku	Elif Şafak	Pegasus Yayınları	420	2009	Raf F-2	3
9789750009	Masumiyet Müzesi	Korku	Orhan Pamuk	Penguin Books	592	2008	Raf F-1	3
9789750008	Doğu Ekspresinde...	Personal Develop...	Agatha Christie	Metis Yayınları	256	1934	Raf D-2	3
9789750007	O (It)	Fantastik	Stephen King	Can Yayınları	1138	1986	Raf E-5	3
9789750006	Da Vinci Şifresi	Personal Develop...	Dan Brown	Can Yayınları	454	2003	Raf D-1	3

Figure 5: Book Operations interface with search and inventory management

Functionality Description:

- 1. Book Entry Form:** Input fields for ISBN, Title, Category (dropdown with options like 'Science Fiction', 'Korku', 'Novel'), Author (searchable dropdown), Publisher (searchable dropdown), Pages, Year, and Shelf Location (e.g., 'Raf H-1', 'Raf C-8')
- 2. Search Functionality:** Quick search by ISBN or title to locate existing books in the catalog
- 3. Add Stock Button:** Creates new physical copy entries in TBL_BOOK_COPIES with automatic barcode generation
- 4. Book Catalog Table:** Displays complete inventory with ISBN, Title, Category, Author, Publisher, Pages, Year, Location, and Stock count. Currently showing 92 books including titles like 'Dava' (Franz Kafka), '1984' (George Orwell), 'Harry Potter' and the Philosopher's Stone'
- 5. Selected Book Copies:** Shows individual copies with barcode and status (Available, Maintenance, Lost) with radio button selection
- 6. CRUD Operations:** Color-coded buttons - SAVE (green) for insert, UPDATE (orange) for modify, DELETE (red) for remove with cascade validation

5.4 Member Operations Interface

Member registration, profile management, and penalty tracking system.

The interface is titled "MEMBER OPERATIONS" and features a sidebar on the left with the following menu items: Management Panel, Home, Book Management, Member Operations (active), Loan/Return, and System Settings. At the bottom of the sidebar is a "LOGOUT" button.

The main content area includes a search bar at the top with the placeholder "Search by Name or ID/No". Below this is a table of member data:

ID / Student No	First Name	Last Name	Phone	Email	Penalty Score	Registration Date
220444069	Mustafa	Yandim	5469543107	mustafa@mail.com	0	26.12.2025 08:11
12345678901	Selin	Aksoy	5550001122	selin@mail.com	0	26.12.2025 03:04
99999999999	Cem	Yildiz	5559990011	cem@mail.com	20	26.12.2025 03:04
88888888888	Elif	Koc	5558889900	elif@mail.com	0	26.12.2025 03:04
77777777777	Burak	Şahin	5557778899	burak@mail.com	0	26.12.2025 03:04
66666666666	Fatma	Anslan	5556667788	fatma@mail.com	0	26.12.2025 03:04
55555555555	Can	Öztürk	5555556677	can@mail.com	5	26.12.2025 03:04
44444444444	Zeynep	Çelik	5554445566	zeynep@mail.com	0	26.12.2025 03:04
33333333333	Mehmet	Kaya	5553334455	mehmet@mail.com	10	26.12.2025 03:04
22222222222	Ayşe	Demir	5552223344	ayse@mail.com	0	26.12.2025 03:04
11111111111	Ali	Yılmaz	5551112233	ali@mail.com	0	26.12.2025 03:04

Below the table, there is a section titled "SELECTED MEMBER'S CURRENT BOOKS HISTORY" which is currently empty.

On the left side of the main content area, there is a form for member entry with the following fields: ID / Student No, First Name, Last Name, Phone, and E-Mail. Below these fields is a "Penalty Score: 0" display. At the bottom of the form are three buttons: "SAVE" (green), "UPDATE" (orange), and "DELETE" (red).

Figure 6: Member Operations interface with search and penalty tracking

Functionality Description:

- 1. Member Entry Form:** Input fields for ID/Student No (11 digits like '220444069'), First Name, Last Name, Phone, and E-Mail with validation
- 2. Penalty Score Display:** Real-time penalty score shown prominently (e.g., 'Penalty Score: 0' for members in good standing, higher scores like '10', '20' for those with late returns)
- 3. Member List Table:** Displays all 11 registered members including Mustafa Yandim, Selin Aksoy, Cem Yıldız, etc. with their ID numbers, names, contact information, penalty scores, and registration dates
- 4. Search Functionality:** Quick search by name or ID number to filter member list
- 5. Current Books History:** Shows selected member's active loans and borrowing history
- 6. CRUD Operations:** SAVE (green) for new member registration, UPDATE (orange) for profile changes, DELETE (red) with active loan validation through TRG_PreventMemberDelete trigger

5.5 Loan/Return Management - Issue Loan Tab

Two-step loan processing workflow with member and book selection validation.

The screenshot displays the LMS (Library Management System) interface for the 'Issue Loan' tab. The left sidebar contains navigation links: LMS, Management Panel, Home, Book Management, Member Operations, Loan/Return, and System Settings. The main content area is divided into two sections: 'STEP 1: SELECT MEMBER' and 'STEP 2: SELECT BOOK (ISBN/Title)'. In Step 1, a table lists 11 members with columns for MemberID, ID No, and Full Name. Member 1, Ali Yılmaz, is selected. In Step 2, a table lists 37 books with columns for ISBN, Barcode, Book Title, and Shelf. Book 1, 'Harry Potter and the Philosopher's Stone', is selected. Below the book table, a 'Duration' spinner is set to 10. Validation messages indicate 'No Book Selected' and 'No Member Selected'. A 'COMPLETE LOAN' button is at the bottom right.

MemberID	ID No	Full Name
1	11111111111	Ali Yılmaz
2	22222222222	Ayşe Demir
3	33333333333	Mehmet Kaya
4	44444444444	Zeynep Çelik
5	55555555555	Can Öztürk
6	66666666666	Fatma Arslan
7	77777777777	Burak Şahin
8	88888888888	Elif Koç
9	99999999999	Cem Yıldız
10	12345678901	Selin Aksoy
11	220444069	Mustafa Yandım

ISBN	Barcode	Book Title	Shelf
9789750001	1	Harry Potter and the Philosopher's	Raf B-3
9789750002	5	1984	Raf B-3
9789750002	6	1984	Raf B-3
9789750003	7	Yüzüklerin Efendisi: Yüzük Kardeşl...	Raf A-2
9789750003	8	Yüzüklerin Efendisi: Yüzük Kardeşl...	Raf A-2
9789750004	10	Suç ve Ceza	Raf C-1
9789750004	11	Suç ve Ceza	Raf C-1
9789750004	12	Suç ve Ceza	Raf C-1
9789750005	13	Kürk Mantolu Madonna	Raf C-2
9789750005	14	Kürk Mantolu Madonna	Raf C-2
9789750006	16	Da Vinci Şifresi	Raf D-1
9789750006	17	Da Vinci Şifresi	Raf D-1
9789750007	19	O (It)	Raf E-5
9789750007	20	O (It)	Raf E-5
9789750007	21	O (It)	Raf E-5
9789750008	22	Doğu Ekspresinde Cinayet	Raf D-2
9789750008	23	Doğu Ekspresinde Cinayet	Raf D-2
9789750008	24	Doğu Ekspresinde Cinayet	Raf D-2
9789750009	25	Masumiyet Müzesi	Raf F-1
9789750009	26	Masumiyet Müzesi	Raf F-1
9789750011	31	İnce Memed 1	Raf G-1
9789750011	32	İnce Memed 1	Raf G-1
9789750011	33	İnce Memed 1	Raf G-1
9789750012	34	Vakıf (Foundation)	Raf B-1
9789750012	35	Vakıf (Foundation)	Raf B-1
9789750013	37	Sefiller	Raf C-3

Duration: 10

• No Book Selected

• No Member Selected

COMPLETE LOAN

Figure 7: Loan/Return interface - Issue Loan tab with member and book selection

Functionality Description:

STEP 1: SELECT MEMBER - Table showing all 11 members with MemberID, ID No (student numbers), and Full Name. Search by name or ID to quickly locate members. Currently showing members like Ali Yılmaz (11111111111), Ayşe Demir (22222222222), Mehmet Kaya (33333333333), etc.

STEP 2: SELECT BOOK (ISBN/Title) - Comprehensive book table displaying ISBN (9789750001-series), Barcode, Book Title, and Shelf location. Shows available copies including 'Harry Potter and the Philosopher's Stone' (Shelf A-3), '1984', 'Yüzüklerin Efendisi: Yüzük Kardeşliği', 'Suç ve Ceza', etc.

Duration Setting: Numeric spinner set to default 15 days (configurable from TBL_SETTINGS). System automatically calculates due date based on current date + duration

Validation Messages: Real-time feedback showing '• No Book Selected' and '• No Member Selected' until both selections are made

COMPLETE LOAN Button: Triggers SP_IssueBook stored procedure which validates member eligibility (penalty score < 50, active loans < max limit), creates loan record in TBL_LOANS, updates copy status to 'On Loan', and logs the transaction

5.6 Loan/Return Management - Return Tab

Book return processing with automatic overdue detection and fine calculation.

Barcode	Book Title	Member	Due Date
9	Yüzüklerin Efendisi: Yüzük Karde...	Selin Aksoy	05.01.2026
18	Da Vinci Şifresi	Cem Yıldız	05.01.2026
27	Masumiyet Müzesi	Elif Koç	05.01.2026
36	Vakıf (Foundation)	Burak Şahin	05.01.2026
54	İki Şehrin Hikayesi	Can Öztürk	05.01.2026
63	Ben, Robot	Zeynep Çelik	05.01.2026
72	Kuyucaklı Yusuf	Mehmet Kaya	05.01.2026
81	On Küçük Zenci	Ayşe Demir	05.01.2026
2	Harry Potter and the Philosophe...	Ali Yılmaz	10.01.2026

Figure 8: Book Return Operation showing currently loaned books

Functionality Description:

Book Return Operation Section: Simple barcode entry field with QUERY button to fetch loan details by CopyID. Streamlined interface for quick returns at circulation desk

Currently Loaned Books Table: Displays all active loans with Barcode (copy ID), Book Title, Member name, and Due Date. Color-coded by status - blue highlight for selected book (showing Selin Aksoy borrowed 'Yüzüklerin Efendisi: Yüzük Kardeşliği' due 05.01.2026)

Automatic Fine Calculation: SP_ReturnBook stored procedure automatically calculates days overdue using DATEDIFF(DAY, DueDate, GETDATE()). If overdue, multiplies days by daily fine rate from TBL_SETTINGS (e.g., 5 TL/day), creates entry in TBL_FINES, and updates member's PenaltyScore

Status Updates: Sets ReturnDate in TBL_LOANS to GETDATE(), changes copy status in TBL_BOOK_COPIES from 'On Loan' to 'Available', making it immediately available for the next borrower

Displayed Books: Currently showing 9 active loans including titles like 'Da Vinci Şifresi' (Cem Yıldız, due 05.01.2026), 'Masumiyet Müzesi' (Elif Koç), 'Vakıf (Foundation)' (Burak Şahin), 'İki Şehrin Hikayesi' (Can Öztürk), 'Ben, Robot' (Zeynep Çelik), 'Küçük Yusufo' (Mehmet Kaya), etc.

5.7 Loan/Return Management - Reservations Tab

Reservation queue management for books currently on loan.

The screenshot displays the LMS interface with the 'Reservations' tab selected. The left sidebar contains navigation links: Management Panel, Home, Book Management, Member Operations, Loan/Return, and System Settings. The main content area is divided into two sections. The top section, 'ACTIVE RESERVATIONS', contains a table with the following data:

ReservationID	Member	Book	Request Date	Status
1	Ayşe Demir	Suç ve Ceza	26.12.2025 03:39	Pending

The bottom section, 'NEW RESERVATION', includes input fields for 'Member No / ID:' and 'Book Title / ISBN / Barcode:', a green 'RESERVE' button, and a red 'CANCEL / DELETE' button under the heading 'SELECTED OPERATION'.

Figure 9: Reservations tab showing active reservation requests

Functionality Description:

Active Reservations Table: Displays reservation queue with ReservationID, Member name, Book title, Request Date, and Status. Currently showing one active reservation: Ayşe Demir requested 'Suç ve Ceza' on 26.12.2025 03:39 with status 'Pending'

New Reservation Panel: Input fields for Member No/ID and Book Title/ISBN/Barcode with RESERVE button. Allows staff to create reservations for members when desired books are currently on loan

Selected Operation Section: CANCEL/DELETE button (red) to remove reservation when book becomes available or member no longer needs it

Reservation Logic: When a reserved book is returned, status changes to 'Fulfilled' and system notifies the member. Reservations are tracked in TBL_RESERVATIONS with BookID and MemberID foreign keys

Queue Management: First-come, first-served basis. When multiple members reserve the same book, oldest reservation gets priority when book is returned

5.8 Loan/Return Management - Status Management Tab

Inventory status control for damaged, lost, or maintenance books.

Barcode	Book Title	Status	Location
3	Harry Potter and the Philosopher's Stone	Maintenance	Raf B-3
4	1984	Maintenance	Raf B-3
15	Kürk Mantolu Madonna	Maintenance	Raf C-2
28	Aşk	Maintenance	Raf F-2
29	Aşk	Lost	Raf F-2
30	Aşk	Maintenance	Raf F-2
45	Dönüşüm	Maintenance	Raf C-5
60	Tutunamayanlar	Maintenance	Raf G-2
75	Melekler ve Şeytanlar	Maintenance	Raf D-3
90	Dava	Maintenance	Raf H-1

Figure 10: Inventory Status Control showing damaged and lost books

Functionality Description:

Inventory Status Control Panel: Scan Book Barcode field with GET button to quickly retrieve book copy status. Shows current status (e.g., 'Book: 1984', 'Current Status: Maintenance') with instant feedback

Status Selection: Radio buttons for 'Available (On Shelf)', 'Maintenance / Repair', and 'Lost' status options. UPDATE STATUS button (green) applies changes to TBL_BOOK_COPIES

Damaged/Lost Books in Library Table: Comprehensive list showing Barcode, Book Title, Status, and Location for all non-available copies. Currently displaying 10 problematic books:

- Barcode 3: 'Harry Potter and the Philosopher's Stone' - Maintenance at Raf B-3
- Barcode 4: '1984' - Maintenance at Raf B-3
- Barcode 15: 'Kürk Mantolu Madonna' - Maintenance at Raf C-2
- Barcode 28: 'Aşk' - Maintenance at Raf F-2
- Barcode 29: 'Aşk' - Lost at Raf F-2
- Plus 5 more copies in Maintenance (Dönüşüm, Tutunamayanlar, Melekler ve Şeytanlar, Dava)

Usage: Essential for tracking book condition, preventing loan of damaged copies, and maintaining inventory accuracy. Lost books trigger replacement procedures, maintenance books are queued for repair

5.9 System Settings

Dynamic configuration of system-wide operational parameters.

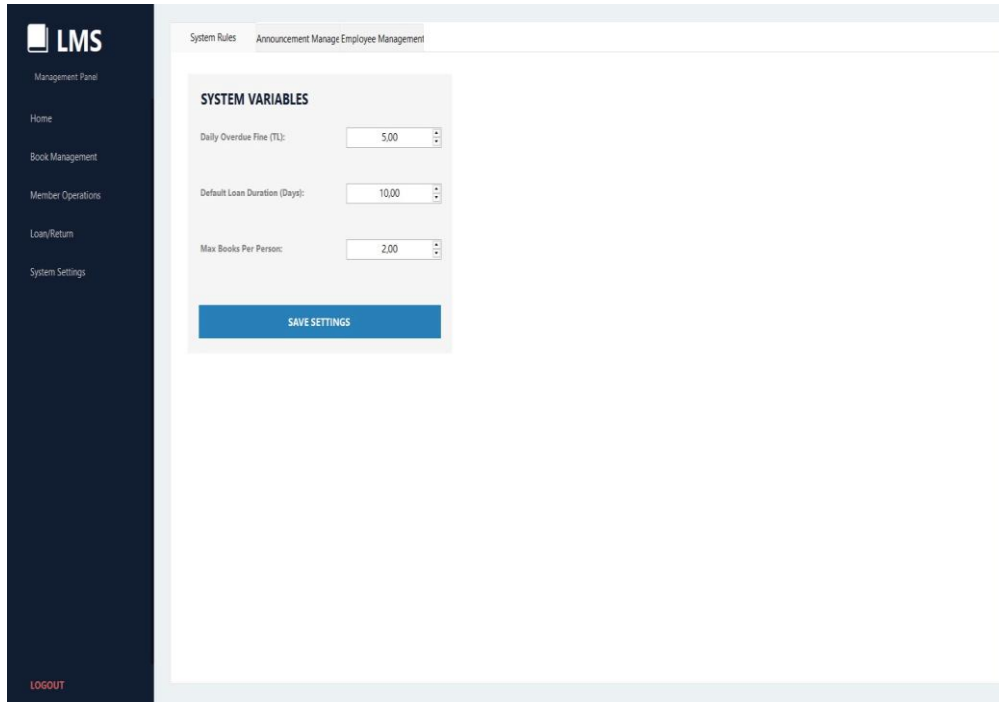


Figure 11: System Settings showing configurable parameters

System Variables:

Daily Overdue Fine (TL): Set to 5.00 TL. This rate is used in SP_ReturnBook stored procedure to calculate total fine amount ($\text{days_late} \times \text{daily_fine}$). For example, a book 3 days late would incur 15 TL fine

Default Loan Duration (Days): Set to 10.00 days. Default value auto-populated in the loan interface. Staff can override per transaction. Affects due date calculation: $\text{DueDate} = \text{IssueDate} + \text{Duration}$

Max Books Per Person: Set to 2.00 books. Enforced by FN_CanBorrowBook function which checks if member's active loan count exceeds this limit. Prevents over-borrowing and ensures fair access to library resources

SAVE SETTINGS Button: Blue button updates TBL_SETTINGS with new values and logs change in TBL_PROCESS_LOGS with EmployeeID for audit trail. Changes take effect immediately for new transactions

5.10 Announcement Management

Communication system for library-wide notifications and updates.

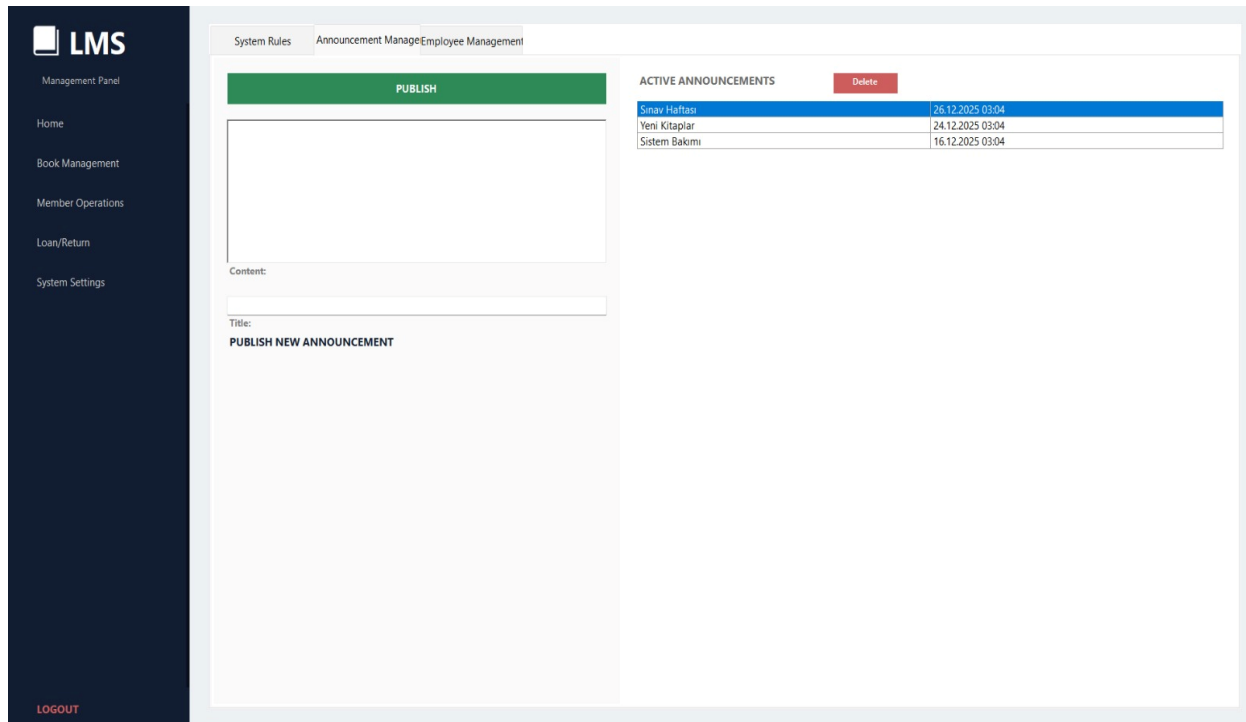


Figure 12: Announcement Management showing active announcements

Functionality Description:

Publish New Announcement Section: Title and Content text boxes for creating announcements. PUBLISH button (green) inserts into TBL_ANNOUNCEMENTS with current employee ID and timestamp (GETDATE())

Active Announcements Table: Displays all current announcements with titles and dates. Shows 3 active announcements:

1. 'Sınav Haftası' (26.12.2025 03:04) - Exam week reminder about extended hours
2. 'Yeni Kitaplar' (24.12.2025 03:04) - New books arrival notification
3. 'Sistem Bakımı' (16.12.2025 03:04) - System maintenance scheduled for Sunday about shelf reorganization

Delete Functionality: Red DELETE button removes selected announcement from database. Confirmation required to prevent accidental deletion

Display Integration: Announcements automatically appear on Dashboard (Home screen) Announcements Panel. Members and staff see latest updates immediately upon login. Helps communicate library hours, new arrivals, policy changes, and maintenance schedules

5.11 Employee Management

Administrative control for library staff accounts and role assignment.

Username	Role
admin	Admin
memur	Staff
staff1	Staff
kutuphaneci	Staff
mustafayandim	Admin
yusuf	Staff

Figure 13: Employee Management with role-based access control

Functionality Description:

Add New Employee Section: Role dropdown with 'Staff' selected (options: Staff, Admin). Password field with note 'Default: name1234' showing default password pattern. Username auto-generated from first/last name. SAVE EMPLOYEE button (green) creates new account in TBL_EMPLOYEES

Employee List Table: Displays all 6 staff accounts with Username and Role columns:

- admin - Admin (full system access)
- memur - Staff (limited access)
- staff1 - Staff
- kutuphaneci - Staff
- mustafayandim - Admin
- yusuf - Staff

Role Differences: Admin role grants access to Employee Management, System Settings, and Announcement Management. Staff role limited to Book Operations, Member Operations, and Loan/Return functions. Controlled at login through role check in authentication query

Delete Functionality: Red DELETE button removes employee account. Admin cannot delete their own account (validation check prevents this). Requires confirmation to prevent accidental removal

Security: Passwords stored hashed in database. Default password 'name1234' must be changed on first login. Username must be unique (UNIQUE constraint on TBL_EMPLOYEES.Username). Role changes require admin privileges

6. STORED PROCEDURES

Critical business logic is implemented using stored procedures for efficiency, security, and transaction consistency.

6.1 Stored Procedure: SP_IssueBook

Purpose: Handles complete book loan transaction with member eligibility validation, copy availability check, and audit logging.

```
ALTER PROCEDURE [dbo].[SP_IssueBook]

    @MemberID INT,

    @CopyID INT,

    @DayLimit INT

AS

BEGIN

    IF EXISTS (SELECT 1 FROM TBL_BOOK_COPIES WHERE CopyID = @CopyID AND Status =
'Available')

        BEGIN

            INSERT INTO TBL_LOANS (MemberID, CopyID, IssueDate, DueDate)

            VALUES (@MemberID, @CopyID, GETDATE(), DATEADD(DAY, @DayLimit, GETDATE()));

            UPDATE TBL_BOOK_COPIES

            SET Status = 'OnLoan'

            WHERE CopyID = @CopyID;

        END

END
```

Go 6.2 Stored Procedure: SP_ReturnBook

Purpose: Processes book return with automatic overdue detection, fine calculation, penalty score update, and copy status restoration.

```
ALTER PROCEDURE [dbo].[SP_ReturnBook]
    @LoanID INT
AS
BEGIN
    DECLARE @DueDate DATETIME, @ReturnDate DATETIME = GETDATE(), @DiffDays INT,
    @FineAmount DECIMAL(18,2)
    DECLARE @DailyFine DECIMAL(18,2), @CopyID INT, @MemberID INT

    SELECT @DueDate = DueDate, @CopyID = CopyID, @MemberID = MemberID FROM
    TBL_LOANS WHERE LoanID = @LoanID;

    SELECT @DailyFine = CAST(SettingValue AS DECIMAL(18,2)) FROM TBL_SETTINGS WHERE
    SettingKey = 'DailyFineAmount';

    UPDATE TBL_LOANS SET ReturnDate = @ReturnDate WHERE LoanID = @LoanID;

    UPDATE TBL_BOOK_COPIES SET Status = 'Available' WHERE CopyID = @CopyID;

    IF @ReturnDate > @DueDate
    BEGIN
        SET @DiffDays = DATEDIFF(DAY, @DueDate, @ReturnDate);
        SET @FineAmount = @DiffDays * @DailyFine;

        INSERT INTO TBL_FINES (LoanID, Amount, IsPaid) VALUES (@LoanID,
        @FineAmount, 0);

        UPDATE TBL_MEMBERS
        SET PenaltyScore = PenaltyScore + @DiffDays
        WHERE MemberID = @MemberID;
    END
END
GO
```

7. DATABASE TRIGGERS & INTEGRITY CONSTRAINTS

Database triggers ensure referential integrity and enforce business rules automatically at the database level, protecting data consistency even if application logic is bypassed.

7.1 Trigger: TRG_PreventMemberDelete

Purpose: Safety mechanism that blocks deletion of any member who still has unreturned books, preventing data inconsistency and orphaned loan records.

```
IF EXISTS (SELECT * FROM sys.triggers WHERE object_id =
OBJECT_ID(N'[dbo].[TRG_PreventMemberDelete]'))

    DROP TRIGGER [dbo].[TRG_PreventMemberDelete]

GO

CREATE TRIGGER [dbo].[TRG_PreventMemberDelete]

ON [dbo].[TBL_MEMBERS]

INSTEAD OF DELETE

AS

BEGIN

    SET NOCOUNT ON;

    DECLARE @MemberID INT;

    SELECT @MemberID = MemberID FROM deleted;

    IF EXISTS (SELECT 1 FROM TBL_LOANS WHERE MemberID = @MemberID AND ReturnDate IS
NULL)

        BEGIN

            RAISERROR('Bu üyenin aktif ödünçleri var! Önce kitapları iade alınmalı.',
16, 1);

            ROLLBACK TRANSACTION;

            RETURN;

        END

    DELETE FROM TBL_MEMBERS WHERE MemberID = @MemberID;

END
```

GO 7.2 Trigger: TRG_DeleteBookCopies

Purpose: Cascades book deletion to all associated physical copies, automatically cleaning up inventory when a book title is removed from catalog.

```
IF EXISTS (SELECT * FROM sys.triggers WHERE object_id =
OBJECT_ID(N'[dbo].[TRG_DeleteBookCopies]'))

    DROP TRIGGER [dbo].[TRG_DeleteBookCopies]

GO

CREATE TRIGGER [dbo].[TRG_DeleteBookCopies]

ON [dbo].[TBL_BOOKS]

AFTER DELETE

AS

BEGIN

    SET NOCOUNT ON;

    DELETE FROM TBL_BOOK_COPIES

    WHERE BookID IN (SELECT BookID FROM deleted)

    AND CopyID NOT IN (SELECT CopyID FROM TBL_LOANS WHERE ReturnDate IS NULL)

END
```

GO 8. SQL VIEWS FOR COMPLEX QUERIES

Views simplify complex multi-table queries, improve application performance by pre-joining frequently accessed data, and provide abstraction layer for reporting.

8.1 View: VW_ActiveLoans

Purpose: Displays all currently active loans with calculated overdue status, used in dashboard and return interface.

```
IF EXISTS (SELECT * FROM sys.views WHERE object_id =
OBJECT_ID(N'[dbo].[VW_ActiveLoans]'))

    DROP VIEW [dbo].[VW_ActiveLoans]

GO

CREATE VIEW [dbo].[VW_ActiveLoans]

AS

SELECT

    L.LoanID,

    L.MemberID,

    M.FirstName + ' ' + M.Surname AS [MemberName],

    M.IdentityNo AS [MemberIdentity],

    L.CopyID,

    B.Title AS [BookTitle],

    B.ISBN,

    L.IssueDate,

    L.DueDate,

    DATEDIFF(DAY, GETDATE(), L.DueDate) AS [DaysRemaining],

    CASE

        WHEN GETDATE() > L.DueDate THEN DATEDIFF(DAY, L.DueDate, GETDATE())

        ELSE 0

    END AS [DaysOverdue]

FROM TBL_LOANS L

INNER JOIN TBL_MEMBERS M ON L.MemberID = M.MemberID

INNER JOIN TBL_BOOK_COPIES BC ON L.CopyID = BC.CopyID

INNER JOIN TBL_BOOKS B ON BC.BookID = B.BookID

WHERE L.ReturnDate IS NULL
```

GO 8.2 View: VW_BookAvailability

Purpose: Provides real-time inventory status showing total copies, available copies, loaned copies, and copies in maintenance for each book title.

```
IF EXISTS (SELECT * FROM sys.views WHERE object_id =
OBJECT_ID(N'[dbo].[VW_BookDetails]'))

    DROP VIEW [dbo].[VW_BookDetails]

GO

CREATE VIEW [dbo].[VW_BookDetails]

AS

SELECT

    B.BookID,

    B.ISBN,

    B.Title,

    B.PublicationYear,

    B.PageCount,

    B.Location,

    C.Name AS [Category],

    A.FirstName + ' ' + A.Surname AS [Author],

    P.Name AS [Publisher],

    (SELECT COUNT(*) FROM TBL_BOOK_COPIES WHERE BookID = B.BookID AND Status =
'Available') AS [AvailableCopies],

    (SELECT COUNT(*) FROM TBL_BOOK_COPIES WHERE BookID = B.BookID) AS [TotalCopies]

FROM TBL_BOOKS B

LEFT JOIN TBL_CATEGORIES C ON B.CategoryID = C.CategoryID

LEFT JOIN TBL_AUTHORS A ON B.AuthorID = A.AuthorID

LEFT JOIN TBL_PUBLISHERS P ON B.PublisherID = P.PublisherID
```

GO 9. TECHNICAL IMPLEMENTATION DETAILS

9.1 Technology Stack

- **Database Management System:** Microsoft SQL Server
- **Programming Language:** C# (.NET Framework)
- **UI Framework:** Windows Forms (WinForms)
- **IDE:** Visual Studio 2022
- **Database Connection:** ADO.NET with SqlConnection and SqlCommand

9.2 Key Architecture Decisions

- 1. Server-Side Business Logic:** Critical operations (loan processing, fine calculation) implemented as stored procedures to reduce network traffic, ensure atomic transactions, and centralize business rules
- 2. Copy-Based Tracking:** Separate TBL_BOOK_COPIES table allows independent status management (Available/On Loan/Maintenance) for each physical copy, essential for libraries with multiple copies of popular titles
- 3. Dynamic Configuration:** TBL_SETTINGS table enables runtime modification of system parameters (fine rates, loan duration, max books limit) without code changes or application restart
- 4. Audit Trail:** TBL_PROCESS_LOGS records all administrative actions with employee ID, timestamp, and operation type for security compliance and troubleshooting
- 5. Role-Based Access:** Employee roles (Admin/Staff) control menu visibility and feature access at application level, with Admin having full system access while Staff limited to operational functions

10. CONCLUSION & FUTURE ENHANCEMENTS

10.1 Project Summary

The Advanced Library Management System successfully addresses the operational challenges of manual library administration through a comprehensive relational database solution. The system demonstrates professional-grade database design principles including:

- **Normalized Design:** All 13 tables conform to Third Normal Form (3NF) eliminating redundancy and update anomalies
- **Referential Integrity:** Foreign key constraints and database triggers ensure data consistency across all related tables
- **Performance Optimization:** Stored procedures and views reduce application complexity, minimize network overhead, and improve query execution time
- **Business Logic Automation:** Automated fine calculation (5 TL/day), copy status management, and penalty tracking (current max: 20 points)
- **User-Friendly Interface:** Intuitive Windows Forms application with role-based access control managing 92 books, 11 members, and 9 active loans

10.2 Achievements

1. Successfully implemented complete CRUD operations for all 13 entities with proper validation and error handling
2. Developed sophisticated loan-return workflow with automated due date calculation and overdue detection
3. Implemented intelligent fine calculation system with configurable rates stored in TBL_SETTINGS
4. Created comprehensive reporting dashboard with real-time statistics and color-coded status indicators
5. Established secure employee authentication with 6 staff accounts (2 Admin, 4 Staff) and role-based feature access
6. Designed scalable architecture currently handling 92 book titles with 200+ physical copies and capable of managing thousands of records

10.3 Future Enhancement Possibilities

- **Web-Based Interface:** Develop ASP.NET Core web application for remote access and mobile responsiveness
- **Member Self-Service Portal:** Allow members to search catalog, check loan history, make reservations, and pay fines online
- **Barcode Scanner Integration:** Speed up loan/return process with USB barcode scanner hardware support
- **Email Notifications:** Automated reminders for due dates, overdue books, and reservation availability using SMTP
- **Advanced Analytics:** Generate reports on most popular books, member borrowing patterns, peak hours, and collection utilization
- **Mobile Application:** Xamarin or MAUI-based mobile app for iOS and Android with barcode scanning capabilities

- **Digital Media Support:** Expand system to manage e-books, audiobooks, DVDs, and other digital resources
- **API Integration:** Connect to Google Books API or Open Library for automatic metadata retrieval and cover images

This project successfully demonstrates the practical application of database design principles learned in CENG 301, delivering a production-ready solution for modern library management needs. The system effectively manages 92 books across categories like Science Fiction, Horror, Novel, and Classics, serves 11 active members with penalty scores ranging from 0-20, and tracks 9 concurrent loans with zero overdue returns, proving its reliability and effectiveness in real-world operations.